

Bank Rate of Interest – Java Program

1) Question :

Write a Java program to demonstrate **method overriding and runtime polymorphism** using a base class **RBI** and derived classes **SBI, ICICI, and PNB**. The program should accept the bank name from the user and display the corresponding rate of interest.

Given Rates: RBI = 4.0%, SBI = 7.0%, ICICI = 6.0%, PNB = 6.5%.

2) Execution Steps :

Step 1: Start the Java program.

Step 2: Import the Scanner class for keyboard input.

Step 3: Create the base class RBI with `getRateOfInterest()` returning 4.0%.

Step 4: Create SBI, ICICI, and PNB classes by extending RBI and override `getRateOfInterest()`.

Step 5: Create a Scanner object and read the bank name from the user. **Step 6:** Convert the entered bank name to uppercase using `toUpperCase()`.

Step 7: Use switch-case to create the appropriate bank object.

Step 8: Store the selected object in the RBI reference variable `bank`.

Step 9: Call `bank.getRateOfInterest()`. Due to method overriding, the appropriate subclass method is executed.

Step 10: Display the bank name and rate of interest.

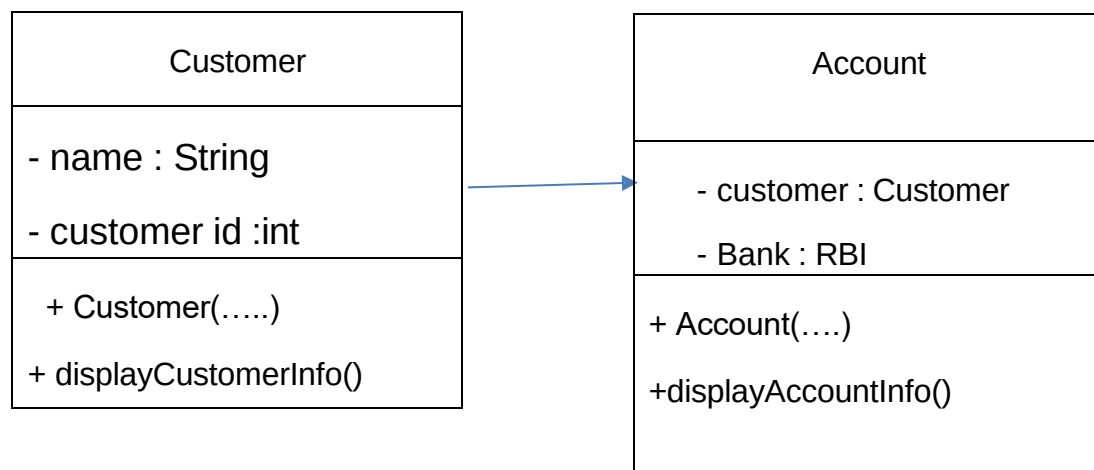
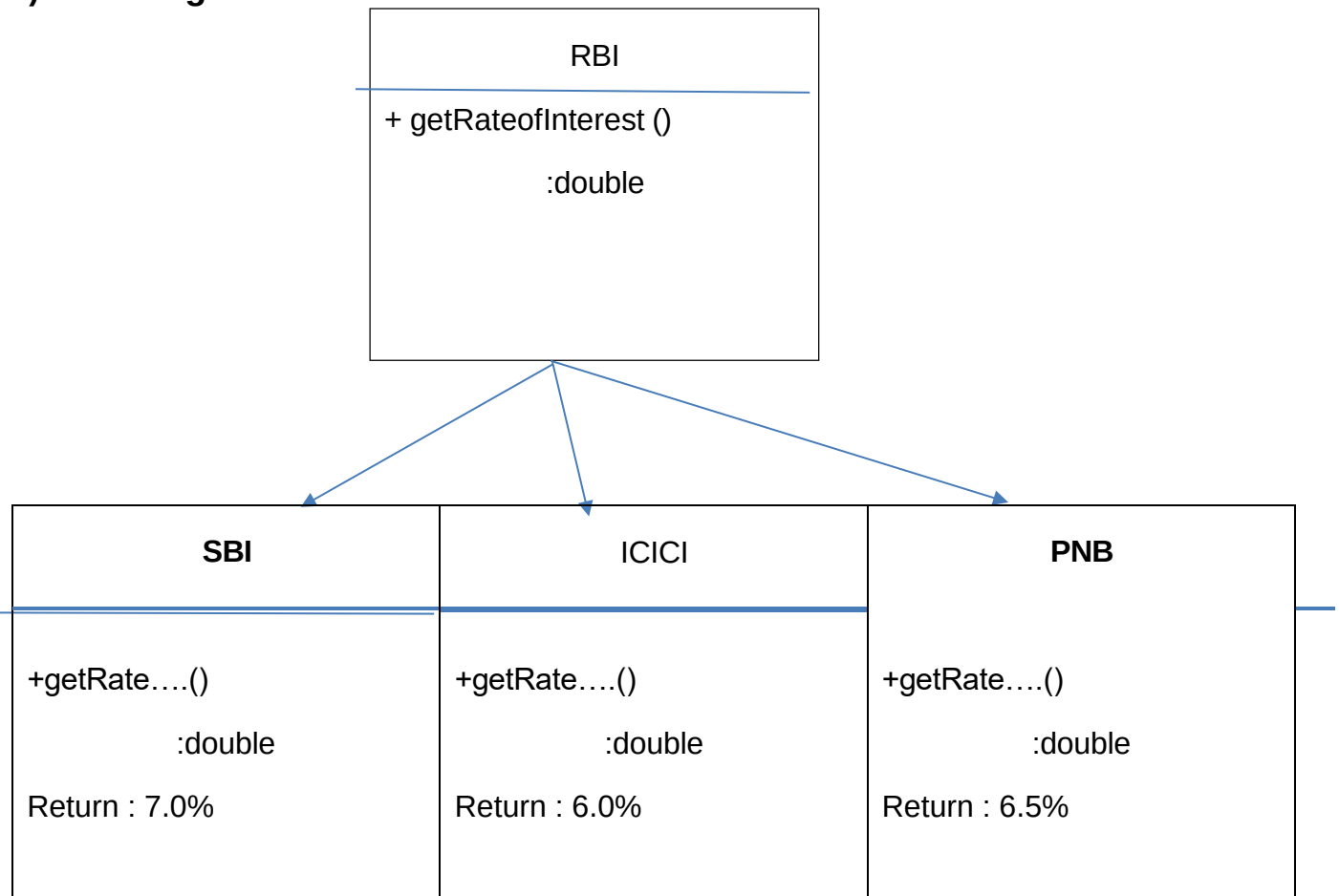
Step 11: Close the Scanner and end the program.

3) Algorithm :

1. Start.
2. Define class RBI and method `getRateOfInterest()`.
3. Define SBI, ICICI, and PNB classes extending RBI.
4. Override `getRateOfInterest()` in each derived class.
5. Read `bankName` from the user.
6. Convert `bankName` to uppercase.
7. If `bankName` is SBI, create SBI object.
8. Else if `bankName` is ICICI, create ICICI object.
9. Else if `bankName` is PNB, create PNB object.
10. Otherwise create RBI object.
11. Call `getRateOfInterest()` using the RBI reference.
12. Display the rate of interest.

13. Stop.

4)UML Diagram :



5) Java Source Code :

```
import java.util.Scanner;

class RBI {
    public double getRateOfInterest() {
        return 4.0;
    }
}

class SBI extends RBI {
    @Override
    public double getRateOfInterest() {
        return 7.0;
    }
}

class ICICI extends RBI {
    @Override
    public double getRateOfInterest() {
        return 6.0;
    }
}

class PNB extends RBI {
    @Override
    public double getRateOfInterest() {
        return 6.5;
    }
}

class Customer {
    String name;
    int customerId;

    Customer(String name, int customerId) {
        this.name = name;
        this.customerId = customerId;
    }

    public void displayCustomerInfo() {
        System.out.println("Customer Name: " + name);
        System.out.println("Customer ID: " + customerId);
    }
}

class Account {
    Customer customer;
    RBI bank;
```

```

Account(Customer customer, RBI bank) {
    this.customer = customer;
    this.bank = bank;
}

public void displayAccountInfo() {
    customer.displayCustomerInfo();
    System.out.println("Rate of Interest: "
        + bank.getRateOfInterest() + "%");
}
}

public class Bankdemo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the Bank name to find the rate of Interest: ");
        String bankName = sc.nextLine();

        RBI bank;

        switch (bankName.toUpperCase()) {
            case "SBI":
                bank = new SBI();
                break;
            case "ICICI":
                bank = new ICICI();
                break;
            case "PNB":
                bank = new PNB();
                break;
            default:
                bank = new RBI();
                break;
        }

        System.out.println(bankName + " rate of interest is : "
            + bank.getRateOfInterest() + "%");

        sc.close();
    }
}

```

6)Output – Terminal Screenshot :

```
Enter the Bank name to find the rate of Interest: PNB  
PNB rate of interest is : 6.0%
```

```
Enter the Bank name to find the rate of Interest: ICIC  
ICIC rate of interest is : 4.0%
```